BEING A MONOMORPHISM IS FPQC LOCAL ON THE BASE (TWO PROOFS: DIAGONAL DESCENT AND REDUCTION TO SCHEMES)

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span contains the author's two labelled arguments 'First proof.' and 'Second proof.', kept verbatim; both are complete derivations and neither is a pointer.

Editorial note. The counted span cites two Topologies on Spaces lemmas, one same-chapter lemma and one Properties of Spaces proposition, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Monomorphism iff diagonal is an isomorphism. Standard, used by the author inside the proof. Supplies the reformulation driving the first proof.
- fpqc coverings are stable under base change. Stacks Project, Topologies on Spaces, Lemma spaces-topologies-lemma-fpqc. Supplies the covering of X \times_Y X used in the descent step.
- fpqc descent of the isomorphism property. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-isomorphism. Supplies that \Delta is an isomorphism when its base changes are.
- Refinement of fpqc coverings by morphisms of schemes. Stacks Project, Topologies on Spaces, Lemma spaces-topologies-lemma-refine-fpqc-schemes. Supplies the reduction of the covering to schemes in the second proof.
- Sheaves form an fpqc sheaf / local equality detection. Stacks Project, Properties of Spaces, Proposition spaces-properties-proposition-sheaf-fpqc. Supplies a = b from agreement over the fpqc covering.

Source reference. Original reference 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041Y: sources/stacks-chow/tags.txt, source lines 5309–5309. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0419: sources/stacks-chow/tags.txt, source lines 5284–5284. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0APL: sources/stacks-chow/tags.txt, source lines 13479–13479. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Lemma 0.1. *The property $\mathcal{P}(f)$ ="f is a monomorphism" is fpqc local on the base.*

Proof. Let $f : X \to Y$ be a morphism of algebraic spaces. Let $\{Y_i \to Y\}$ be an fpqc covering, and assume each of the base changes $f_i : X_i \to Y_i$ of f is a monomorphism. We have to show that f is a monomorphism.

First proof. Note that f is a monomorphism if and only if $\Delta : X \to X \times_Y X$ is an isomorphism. By applying this to f_i we see that each of the morphisms

$$\Delta_i : X_i \longrightarrow X_i \times_{Y_i} X_i = Y_i \times_Y (X \times_Y X)$$

is an isomorphism. The base change of an fpqc covering is an fpqc covering, see Topologies on Spaces, Lemma 03MR hence $\{Y_i \times_Y (X \times_Y X) \to X \times_Y X\}$ is an fpqc covering of algebraic spaces. Moreover, each Δ_i is the base change of the morphism $\Delta : X \to X \times_Y X$. Hence it follows from Lemma 041Y that Δ is an isomorphism, i.e., f is a monomorphism.

Second proof. Let V be a scheme, and let $V \to Y$ be a surjective étale morphism. If we can show that $V \times_Y X \to V$ is a monomorphism, then it follows that $X \to Y$ is a monomorphism. Namely, given any cartesian diagram of sheaves

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{a} & \mathcal{G} \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle c} \\ \mathcal{H} & \xrightarrow{d} & \mathcal{I} \end{array} \qquad \mathcal{F} = \mathcal{H} \times_{\mathcal{I}} \mathcal{G}$$

if c is a surjection of sheaves, and a is injective, then also d is injective. This reduces the problem to the case where Y is a scheme. Moreover, in this case we may assume that the algebraic spaces Y_i are schemes also, since we can always refine the covering to place ourselves in this situation, see Topologies on Spaces, Lemma 0419.

Assume $\{Y_i \to Y\}$ is an fpqc covering of schemes. Let $a, b : T \to X$ be two morphisms such that $f \circ a = f \circ b$. We have to show that $a = b$. Since f_i is a monomorphism we see that $a_i = b_i$, where $a_i, b_i : Y_i \times_Y T \to X_i$ are the base changes. In particular the compositions $Y_i \times_Y T \to T \to X$ are equal. Since $\{Y_i \times_Y T \to T\}$ is an fpqc covering we deduce that $a = b$ from Properties of Spaces, Proposition 0APL. □